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Paper 2

Mark scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from [oxfordaqaexams.org.uk](https://www.oxfordaqaexams.org.uk)

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	electron		1	AO1 3.1.2 j

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	neutron		1	AO1 3.1.2 d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	there should be one less proton or there should be four protons	allow shell for energy level	1	AO2 × 2 AO3 × 2 3.1.2 f h i
	there should be one less neutron or there should be five neutrons		1	
	any two from: - there should be two electrons on the inner energy level (not one) - there should be two electrons on the outer energy level - there should be four electrons in total		2	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	transition metals		1	AO2 3.7.2 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	$(M_r =) 56 + (16 \times 3) + (1 \times 3)$ $= 107$		1 1	AO2 3.6.2 a
Total Question 1			9	

Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	iodine		1	AO2 3.7.1 d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	labelled bar for bromine at 0.11 (nm)	allow a tolerance of $\pm$ half a small square	1	AO2 3.7.1 c
	labelled bar for iodine at 0.13 (nm)	if no mark awarded allow 1 mark for correctly plotted bromine and iodine bars.	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	sodium iodide solution		1	AO2 3.7.1 d e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	-1		1	AO1 3.2.1 e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	2		1	AO2 3.2.1 c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	giant (structure)	allow (ionic) lattice allow regular structure	1	AO1 3.2.1 f 3.2.2 a
	electrostatic forces of attraction		1	
	between oppositely charged ions		1	
	(forces) in all directions		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	D		1	AO3 3.2.2 a b

Total Question 2		11
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	C ₇ H ₁₄		1	AO2 3.10.1 2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	contains a double (covalent) bond (between two carbon atoms)		1	AO1 3.10.1.3 c

Question	Answers	Mark	AO/ Spec. Ref.
03.3	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3–4	AO3 3.10.2 d
	Level 1: Relevant points are made. They are not logically linked.	1–2	
	No relevant content	0	
	Indicative content: • a reactive polymer could react in the mouth • polymer F not suitable because it is reactive • but polymers E and G are suitable because they are unreactive • low hardness polymer would wear away due to repeated use • polymer E not suitable as low hardness • but polymers F and G are suitable as high hardness • water soluble polymers would dissolve in the mouth • polymer E not suitable as soluble in water • but polymers F and G are suitable as insoluble in water • polymer G is more expensive than polymer E and F • overall polymer G has most suitable properties but it is expensive.		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	(volume absorbed =) 4 cm ³	allow correct use of an incorrectly determined volume absorbed	1	AO2 × 1 AO3 × 1 3.10 2 d 3.5.1 f
	$\left(\% \text{ absorbed} = \frac{4}{50} \times 100 = \right)$ 8(%)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	C–C and trailing bonds	an answer of: $n \begin{array}{c} \text{H} \quad \text{OH} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array} \longrightarrow \left(\begin{array}{c} \text{H} \quad \text{OH} \\ \quad \\ -\text{C} - \text{C}- \\ \quad \\ \text{H} \quad \text{H} \end{array} \right)_n$	1	AO2 3.10. 2a
	OH and three Hs attached to two Cs		1	
	brackets and n		1	

 scores **3** marks

Total Question 3	11
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	(material) glass or metal	allow named metal	1	AO4 3.9.2 a
	(reason) any one from: - metal is a conductor of heat - glass / metal will not melt - glass / metal will not burn in the flame	allow plastic would melt allow plastic would burn in the flame	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	(the experimental amount of) energy released does not double when the number of carbon atoms doubles.	allow on a graph the line of best fit would not go through the origin	1	AO3 3.9.2 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	not all the energy released is transferred to the water	allow because energy is lost to the surroundings allow because heat is lost to the surroundings	1	AO3 3.9.2 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	(temperature change = 35.5 – 20.2 =) 15.3 (°C)		1	AO2 3.9.2 a b
	(Q = 15.3 × 4.2 × 100 =) 6426 (J)	allow correct use of incorrectly calculated temperature change	1	
	(mass burned = 80.52 – 80.16 =) 0.36 (g)		1	
	$\left(\text{energy per g} = \frac{6426}{0.36} = \right)$ 17 850 (J/g)	allow correct use of incorrectly calculated mass burned	1	
	= 17.85 (kJ/g)	allow correct conversion of incorrectly calculated energy per gram	1	
Total Question 4			9	

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	(gold is) unreactive		1	AO1 3.3.1.1 c

Question	Answers	Mark	AO/ Spec. Ref.
05.2	Level 2: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	3–4	AO3 3.3.1 f
	Level 1: Some logically linked reasons are given. There may also be a simple judgement.	1–2	
	No relevant content	0	
	Indicative content: - plant A absorbs copper and nickel which could make extracting the copper more difficult - if both ions present in plant A were extracted could make extraction more valuable - plant A produces 129 mg of copper per 100 m² area - plant B produces the smallest amount of copper in mg/kg per plant - but plant B has highest plant growth per 100 m² area - plant B produces 130 mg of copper per 100 m² area - plant C produces the largest amount of copper in mg/kg per plant - but plant C has lowest plant growth per 100 m² area - plant C produces 106 mg of copper per 100 m² area - plant B therefore produces the highest mass of copper per 100 m² area.		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	increase the surface area (of the ore)	allow increase the rate of reaction	1	AO4 3.8.1 f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	mass of copper produced increases with increasing temperature	allow a value in the range 41 – 44 °C	1	AO3 3.3.1 f
	up to 42 °C		1	
	mass of copper produced decreases with increasing temperature (above 42 °C)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	(mole of copper =) $\left(\frac{381}{63.5} =\right) 6$	allow correct use of incorrectly calculated value of moles	1	AO2 3.6.1 d
	(mass of iron =) 6×56		1	
	= 336 (g)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.6	(metals contain) delocalised electrons	ignore (which are) free to move throughout the (whole) structure / copper	1	AO1 3.2.1 j 3.2.2 g
	(which are) free to move through the (whole) structure / copper		1	

Total Question 5	14
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	(hydrogen) bubbles / gas given off more slowly		1	AO3 3.3.1.1 a
	temperature increases by less (than 12 °C)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	magnesium chloride	allow MgCl_2	1	AO2 3.5.2 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	(test) lighted splint	MP2 is dependent on MP1 being awarded	1	AO1 3.4.2 a
	(result) makes a pop (sound)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	potassium is too reactive	allow too dangerous	1	AO2 3.5.1 c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	$2 \text{H}_3\text{PO}_4 + 3 \text{Mg} \rightarrow \text{Mg}_3(\text{PO}_4)_2 + 3 \text{H}_2$	allow multiples	1	AO2 3.6.1 a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.6	mass of hydrogen is low so mass loss would be small		1	AO4 3.3.1.1 a

Total Question 6		8
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Question 7

Question	Answers		Mark	AO/ Spec. Ref.
07.1	Metal ion Barium Potassium Flame colour Crimson Green Lilac Red Yellow do not accept more than one line from a box on the left		1 1	AO1 3.4.3 a

Question	Answers	Mark	AO/ Spec. Ref.
07.2	Level 3: The method would lead to the production of a valid outcome. The key steps are identified and logically sequenced.	5–6	AO1 × 4 AO4 × 2 3.4.3 b c e
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content: • add sodium hydroxide solution • followed by excess sodium hydroxide solution • to a separate sample add (acidified) silver nitrate solution aluminium chloride • will give a white precipitate on addition of sodium hydroxide solution • which will dissolve in excess sodium hydroxide solution • will give a white precipitate on addition of (acidified) silver nitrate solution calcium chloride • will give a white precipitate on addition of sodium hydroxide solution • which will not dissolve in excess sodium hydroxide solution • will give a white precipitate on addition of (acidified) silver nitrate solution magnesium bromide • will give a white precipitate on addition of sodium hydroxide solution • which will not dissolve in excess sodium hydroxide solution • will give a cream precipitate on addition of (acidified) silver nitrate copper bromide • will give a blue precipitate on addition of sodium hydroxide solution • which will not dissolve in excess sodium hydroxide solution • will give a cream precipitate on addition of (acidified) silver nitrate.		
Total Question 7		8	

Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	lead nitrate		1	AO3 3.5.2 c
	any soluble chloride	do not accept lead / silver chloride	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	filter the mixture		1	AO4 3.4.1 c
	wash the residue		1	
	dry the filter paper	allow dry the residue allow dry the lead chloride	1	
	weigh the precipitate	allow weigh the (dry) residue allow weigh the (dry) lead chloride	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	28 (cm ³)		1	AO3 3.5.2 c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	no ions in solution		1	AO2 3.2.2 b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.5	(conversion of 50 cm ³) = 0.050 dm ³		1	AO2 3.6.3 a 3.6.4 a
	(moles of Ba(OH) ₂ = 0.05 × 0.80) = 0.040 (mol)	allow correct use of incorrectly / not converted value for volume	1	
	(mass of Ba(OH) ₂ =) 0.04 × 171	allow correct use of incorrectly calculated value for moles	1	
	= 6.84 g		1	
Total Question 8			12	

Question 9

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.1	the reaction is reversible		1	AO3 3.6.1 e 3.9.1 d
	(some of the ethanol / gas) may be lost (when it is separated from the reaction mixture)		1	
	(some of the ethene) has reacted in a different way to expected (to produce a white solid)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.2	(as pressure increases) the equilibrium position shifts to the left hand side		1	AO2 3.8.2 d
	(because) there are fewer moles / molecules of gas on the left hand side		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.3	the percentage yield of product / hydrogen / H ₂ increases (as temperature increases)		1	AO2 × 2 AO3 × 1 3.8.2 c
	(because the forward) reaction is endothermic or (because) the backward reaction is exothermic		1	
	(so the) equilibrium position shifts to right hand side		1	

Total Question 9		8
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